

Augmenting VR Ski Training using Time Distortion

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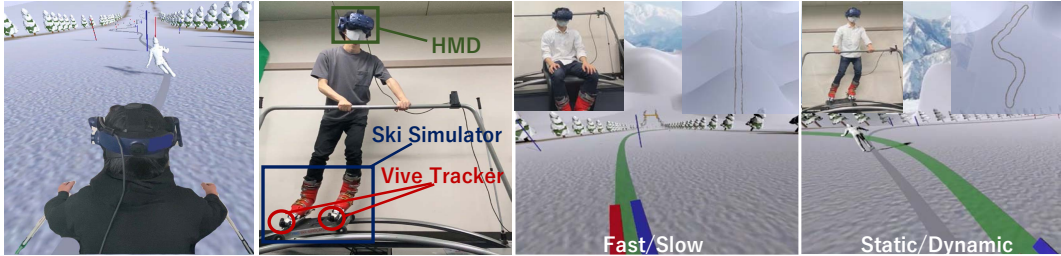


Figure 1: Overview of the system and the scene of two experiments to study time distortion.

ABSTRACT

Virtual reality-based sports simulators are widely developed, which makes training in a virtual environment possible. On the other hand, methods using temporal features are also introduced to realize an adaptive training. In this paper, we study the effect of time distortion on alpine ski training to find out how modifying the temporal space can affect sports training. Experiments are conducted to investigate how a fast/slow and a static/dynamic time distortion-based training, respectively, can impact the performance of users.

Index Terms:

Human-centered computing—Virtual reality

1 INTRODUCTION

Training is always the dominant way to improve one's skill, especially for competitive sports. However, not all the sports can be trained easily, one example is the alpine skiing slalom, where the skier must pass alternating pairs of red and blue poles with very small offsets. The training is difficult to perform because of the requirements of specific equipment and environment. Even the National Olympic ski team of the United States is using VR HMD for training in pre-recorded race courses.¹ A great number of researches are conducted on combining ski simulators with VR to support the training [3–5], but most of them are made for intermediate or higher level skier, both the difficulty and their visual instructions are too informative for beginner. Many recent works introduce several time distortion (or temporal modification) in VR, such as slow motion, fast motion, or dynamic temporal changes, to support training, which is helpful in reducing the huge gap between novice and expert.

Therefore, in this paper, we propose a new VR ski training system augmented by time distortion and study the effect of different temporal modification for alpine skiing slalom. There are several important parameters in time distortion, we focus on the speed (fast/slow) and the way of modification (static/dynamic). A speed-up time space can provide more training and help the trainee to get used to the speed while a speed-down condition can provide more time for thinking and observing. On the other hand, static (constant speed) is the most common time distortion while dynamic (adaptive speed) condition

can modify the time space based on specific target (such as trainee's performance or the difficulty of skill). Two experiments are held to study different temporal parameters in VR skiing, as shown in Figure 1, to find out which type of time distortion is most suitable for VR ski training in terms of improving performance and usability.

2 RELATED WORKS

Nozawa et al. [4] are the predecessors who combine a simplified motion ski simulator with VR training. Motion data of an expert skier is replayed in front of the user as a coach in VR. However, they do not perform any study on the training effect of different visualizations. Wu et al. [5] extended Nozawa et al.'s system by developing seven different visualization feedback methods to support the user. Nevertheless, their results suggest that complex visualization, such as plotting the angle of both feet on a graph, is less effective for beginners because they cannot handle the information well.

Kawasaki et al. [2] proposed a system that allows users to practice kendama skills (a traditional Japanese game) by slowing down the kendama ball and then gradually increasing the speed to normal. Their experiments showed that most of the system's users were able to learn faster than those who trained in the real world. Hamanishi et al. [1] introduce a dynamic time distortion in VR tennis by controlling the speed of time space according to the user's movement. All these previous researches of time distortion show the possibility to make a smooth transition for beginners which leads to an improvement in training performance.

3 SYSTEM DESIGN

Hardware System: The simulation system developed in this paper consists of an indoor ski simulator²(PRO-SKI SIMULATOR), an HMD with 90Hz refresh rate(HTC Vive Pro), and two Vive trackers as shown in the second figure in Figure 1. By attaching the Vive trackers to the tips of the skis, we can obtain the rotation and position of the skis.

Skiing Control: We created a virtual ski slope environment to measure the ski performance. Since the motion simulator used in our setup only has 3 degree of freedom, to simplify the operation, the direction control is defined as a circular curve motion in the horizontal plane. Here, the radius R of the turning circle is defined as $R = 100 - 85 \left(\frac{\min(20, \frac{a_l + a_r}{2})}{20} \right)$, where a_l and a_r are the inclination angles of the left and right ski, respectively.

Replay Function: In order to support the user's training, we implement a replay function that lets the user to view a replay of an expert's run from a first-person perspective, and a leading coach

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¹<https://usskiandsnowboard.org/news/us-ski-snowboard-and-strivr>

²<https://www.ski-simulator.com/product/power-ski-simulator/>

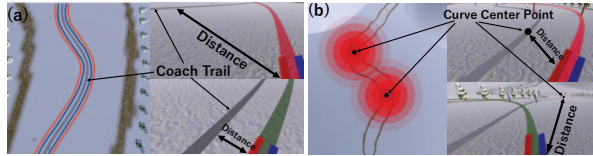


Figure 2: Two dynamic time distortion conditions.

function which replays an expert's full body motion before the user. The temporal replay speed can be changed according to users' needs.

3.1 Time Distortion Functions:

We develop three types of time distortion method to change the temporal space of the skiing, a static conditions using a constant magnification M ($M0.8$ means $0.8x$ slow down motion) to study the effect of speed rate. Besides, two dynamic conditions where the time is distorted by referring to different targets (Distance to Coach or distance to Curve Points, as shown in Figure 2(a)).

For the Distance to Coach function, the speed of time is changed according to the distance D in the last 100 frames (about 1 sec), where D refers to the minimum distance between the user and the trail of coach. According to suggestions from experienced skier, error within 1 meter ($D < 1.0$) is considered to be a good performance and will increase the speed, while error greater than 1.5 meter ($D > 1.5$) will decrease it.

Similarly, the Distance to Curve Point function is developed because it is considered that large turns in slalom are the most difficult part. This time, D refers to the distance between the user and the closest curve center point. In our prototype, whenever the distance decreases 25 meter, the speed increases by 0.1 times which range between $0.5x$ to $1.0x$.

4 USER STUDY

In order to verify a better method of time distortion, two experiments were conducted.

In the first study, we examined whether a speed-up or speed-down condition benefits more for beginners using qualitative interviews. Data are collected from 6 participants (5 males, avg. Age=24.0), where all of them are novice in alpine skiing. As a result, the user's preference suggest that both the fast and the slow time distortions are better than a base normal speed condition while the slow motion is significantly better than the others for beginners.

In the second study, we investigate the slow down time distortion one step further. A baseline condition ($1.0x$), two static conditions ($0.6x$, $0.8x$) together with the two dynamic conditions are studied. The performance (number of successfully passed poles) of before and after each practicing is used for evaluations while another questionnaire is also given to the users. Finally, we collect data from 12 novice participants (9 males, avg. Age = 23.6). The quantitative result is shown in Figure 3. Users' performance are significantly improved compared in all the time distortion conditions compared to the baseline. An average of approx. 15 poles are passed after training in the 4 time distortion conditions which shows a 36% improvement from the baseline. Although there is no significant difference between each time distortion conditions, the users tend to prefer the static condition ($0.6x/0.8x$) more which is indicated by the questionnaire. We obtain comments such as "It was difficult to respond to the speed change during the run".

5 DISCUSSION

First of all, compared to no time modification (normal time speed), it is effective to perform time distortion for beginners to provide a smoother transition to high-level skill. When comparing slow and fast conditions, it is suggested to use a slow one which can observe the coach's motion more clearly than to get used to a faster speed.

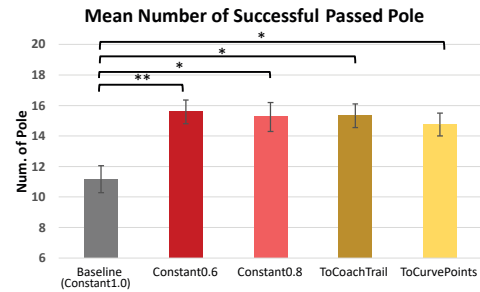


Figure 3: Mean number of successful pass through the poles after each training. Brackets indicate significant differences.

Based on the current results, dynamic speed changes does not necessarily provide benefits and therefore needs to be carefully considered. However, this only applies for the methods introduced in this paper, as suggested by some participants, a time distortion after each trial may be more effective than changes during races. Therefore, a more detailed study with more participants is required.

Overall, this work still has several limitations which can be fixed in the future: First, the ski simulator used in this paper is a rather simple motion simulator, and the physics simulation is also prototype which can be improved. Second, the dynamic conditions are not preferred by some participants, there is some feedback suggesting to change the speed after each trial instead of after several frames during the race. Lastly, in the current system, when the user and the coach is too far away, it is difficult to see the coach, and imitation becomes impossible. This problem may be solved by displaying the coach not only on the slope but also at a fixed position on the HMD.

6 CONCLUSION

In this paper, we introduced a VR training system that combines alpine skiing with time distortion functions. We compare the training effect of several time distortion algorithms with a baseline condition. The results suggest that time distortion can improve skiing just as other sports. Even though we are not able to find out any significant difference between each time distortion methods, it shows a tendency that a static time distortion is more likely to be preferred by the users, while a dynamic one during a run is less likely to be adapted.

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